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Daniel McFadden and the Econometric Analysis of Discrete Choice

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I. Introduction

This article describes the contributions of Daniel McFadden to the econometric analysis of discrete choice. Section I overviews the main elements of McFadden's research on discrete choice analysis, and notes some of his other contributions to economics. Section II addresses the fundamental methodological breakthrough that McFadden made in "Conditional Logit Analysis of Qualitative Choice Behavior" [III, 1974].[1] Section III discusses the subsequent development of discrete choice analysis into one of the foundations of modern econometrics. Section IV concludes.

An Overview of McFadden's Contributions to Discrete Choice Analysis

In 1968 Daniel McFadden circulated a working paper "Revealed Preferences of a Government Bureaucracy," eventually published as McFadden [III, 1975a and III, 1976a]. In this empirical study of the decision-making of the California Highway Commission, McFadden introduced the set of methodological ideas that developed into the modern econometric analysis of discrete choice. In the early 1970s, he deepened and refined his ideas as he performed an influential empirical study of household travel decisions, eventually published as *Urban Travel Demand: A Behavioral Analysis* by Domencich and McFadden [1975]. At the same time, he abstracted from particular empirical questions to consider in generality the problem of econometric inference on individual behavior. The result was McFadden's classic contribution to econometric theory, "Conditional Logit Analysis of Qualitative Choice Behavior" [III, 1974]. The close coordination between development of econometric theory in this article and its application to empirical problems of substantive importance is emblematic of McFadden's research

*I have benefited from the comments of Joel Horowitz and Gadi Barlevy.

[1]Dates in brackets refer to publications by McFadden (see Bibliography). Those preceded by a Roman numeral refer to the subject headings of articles written by McFadden; those without a Roman numeral refer to his books.

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throughout his career. Theoretical development and empirical application nurture each other, yielding contributions of greater significance than would each in isolation.

Discrete choice analysis developed very rapidly during the 1970s as a group of econometricians and empirical researchers joined McFadden in generalizing the econometric theory and applying it to a wide range of applications. In addition to making major contributions of his own during this period, McFadden did much to encourage the emergence of a coherent research community by initiating a highly stimulating series of conferences on the structural analysis of discrete data. These conferences yielded two edited volumes that helped to spread interest among economists and more widely; see *Annals of Economic and Social Measurement* (1976)[2] and Manski and McFadden [1981].

One objective of McFadden's own methodological research in the period was to loosen the behavioral restrictions of the conditional logit model, while maintaining computational tractability. This work culminated in development of the *generalized extreme value* model in McFadden [III, 1981a]. Another objective, also expressed in McFadden [III, 1981a], was to integrate the analysis of discrete choice with the classical theory of consumer demand. A third focus was collaborative research on the estimation of discrete choice models from choice-based samples; see Manski and McFadden [III, 1981b].

McFadden's applied research in the 1970s was channeled primarily through the Urban Travel Demand Forecasting Project that he headed at the University of California, Berkeley; see McFadden *et al.* [II, 1979]. The project was a major incubator and testing ground for new ideas. Moreover, it trained a generation of Berkeley Ph.D. students in the art of applied econometric analysis.

Throughout most of the 1980s, McFadden's research emphasized the application of discrete choice analysis rather than new methodological development. His research in the first half of the decade focused on residential energy choices, as in Cowing and McFadden [1984], and then shifted to the demand for residential telephone service, as in McFadden, Train, and Ben-Akiva [III, 1987a]), and the housing choices of the elderly, as in McFadden [VII, 1994a]). The work on energy choices made an important methodological contribution to the analysis of choice problems having both discrete and continuous components. The resulting article, Dubin and McFadden [VI, 1984a], was awarded the Econometric Society's Frisch Medal in 1986.

Publication of "A Method of Simulated Moments for Estimation of Discrete Response Models without Numerical Integration" [III, 1989a] marked McFadden's re-emergence as a major force in methodological

[2]Dates in parentheses refer to works by other authors listed at the end of this article.

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research. Since then, he has further developed the method of simulated moments and has studied related approaches using simulation as a computational aid in estimation.

McFadden's Contributions to Other Fields of Economics

The present article limits itself to describing McFadden's contributions to discrete choice analysis. Nevertheless, it would be remiss not to take note of his contributions to several other fields of economics. In the 1960s and early 1970s, McFadden worked intensively on the theoretical and econometric analysis of production. During this period he also contributed to economic growth theory, welfare economics, and mathematical economics. In the 1990s, he wrote a series of articles on the contingent valuation of environmental goods.

McFadden's early research on production analysis exhibits the same philosophy that proved so successful in the development of discrete choice analysis. Care is taken to propose econometric methods that are consistent with basic ideas in economic theory. The development of econometric method goes hand in hand with empirical application. Although McFadden's research on production analysis was performed largely in the 1960s, most of his contributions remained unpublished until the volume edited by Fuss and McFadden [1978] appeared in print.

II. Conditional Logit Analysis of Qualitative Choice Behavior

The economic conceptualization of decision-making as maximization of utility subject to constraints has exceedingly broad potential application. However, the empirical economic literature of 30 years ago was largely concerned with the limited class of decision problems envisioned in classical consumer demand theory, wherein the objects of choice are bundles of homogeneous and infinitely divisible goods available to consumers at given prices per unit. To the extent that choice problems with other characteristics were examined, the prevailing practice was to shoehorn them into the classical framework. This resulted in studies of "travel demand" or the "demand for education" or the "demand for consumer durables."

The econometric analysis of discrete choice developed in McFadden [III, 1974] was intended to provide an appropriate framework for the empirical analysis of choice among finite sets of alternatives, with each alternative characterized as a bundle of attributes. McFadden's vision of an appropriate framework had four essential features:

(i) Econometric analysis of behavior should be fully consistent with utility theory. In particular, econometric analysis should be formally

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interpretable as describing the behavior of a population of heterogeneous decision-makers, each of whom chooses the best available alternative.

(ii) Econometric analysis should enable the researcher to forecast choice behavior in new settings. In particular, econometric analysis should enable the researcher to perform thought experiments asking how population behavior would change if new alternatives were to become available or existing ones were to become unavailable.

(iii) Econometric analysis should recognize that the empirical researcher typically has incomplete data on the attributes of the decision-makers that compose the population and the alternatives available to them.

(iv) Econometric analysis should be computationally practical.

The challenge was to embody these features in a framework for the econometric analysis of discrete choice. McFadden's solution is described below.

(i) Consistency with Utility Theory through Random Utility Models

Applying basic utility theory to the problem of discrete choice, McFadden supposed that each member of a population of interest faces a finite choice set and selects an alternative that maximizes utility. Suppose that person t faces choice set C_t, has utility function $u_t(\cdot)$ assigning a value to each potentially available alternative, and chooses some alternative $i \in C_t$. Then the standard revealed preference argument holds that $u_t(i) \geqslant u_t(j)$, all $j \in C_t$.

Assume that alternative i is nominally available to all members of the population and that a negligible fraction of the population are indifferent between i and other available alternatives. Then the fraction of the population who choose i equals the fraction of the population for whom i is the utility maximizing alternative. Formally, the *choice probability* for alternative i is

$$P(i) = P[t\colon u_t(i) \geqslant u_t(j), \text{ all } j \in C_t]. \tag{1}$$

The RHS of (1) is the probability that a person chosen at random from the population has a utility function that makes i the utility maximizing alternative. Equation (1) is referred to as a *random utility model* (RUM) explanation of observed choices.

In psychology, random utility models date back as far as Thurstone (1927) as a way of conceptualizing semi-rational behavior. The psychological interpretation, exposited in Luce and Suppes (1965), assumes that each decision-maker carries a distribution of utility functions internally and

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selects one at random whenever a decision must be made. McFadden's simple insight was to re-interpret the randomness as arising from cross-sectional variation in utility functions across the population rather than from time-series variation within a given individual.

McFadden was the first to recognize the broad power of the RUM as a tool in econometric analysis, but he was careful to acknowledge connections with other work in economics. McFadden [III, 1974, footnote 7] notes that the RUM idea is implicit in random coefficient models of classical consumer demand. He also calls attention to the independent use of the idea in a discrete choice application by Quandt (e.g. 1970).

(ii) Forecasting in New Settings through Attribute Characterizations of Alternatives and Decision-makers

In its abstract form, the random utility model can describe observed population behavior but does not provide a way to forecast behavior in new settings. An abstract RUM cannot be used to predict how a given decision-maker would behave when facing a different set of alternatives. Nor can it be used to predict the behavior of decision-makers in a different population.

McFadden saw that forecasting is enabled if the qualitative distinctions among alternatives and among decision-makers can be mapped into quantitative differences in their attributes. Thus the second step in the development of discrete choice analysis was to assume that each alternative j can be characterized through a vector of attributes x_j and that each decision-maker t can be characterized through a vector of attributes s_t. Then the utility of alternative j to person t has the form

$$u_t(j) = u(x_j, s_t), \tag{2}$$

where $u(\cdot, \cdot)$ maps the attributes of alternatives and decision-makers into utility values. Given knowledge of these attributes and of the form of $u(\cdot, \cdot)$, a researcher can determine the utility of any alternative to any decision-maker and, hence, can predict the decision-maker's choice behavior. For example, a transportation researcher can predict how commuters with specified income and job attributes would choose among travel modes with specified travel-time and travel-cost attributes.

McFadden's explicit characterization of alternatives and decision-makers as attribute vectors marked a sharp departure from prevailing economic practice. In classical consumer demand, goods are qualitatively distinct and so are consumers. There is no way to predict the demand for a new good. Nor is there any way to predict the behavior of new consumers.

A notable exception to the standard practice was the work of Lancaster (1966), which did recognize the value of attribute characterizations.

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Lancaster sought to extend classical demand theory by recasting it as a theory of the demand for attributes. In his approach, attributes are infinitely divisible and available to consumers at a given price per unit. Consumers choose among a continuum of attribute bundles subject to a budget constraint.

(iii) Analysis with Incomplete Data through Conditional Choice Probabilities

One more conceptual step was needed to yield a viable approach to discrete choice analysis. It is not realistic to think that an empirical researcher will have complete data on the attributes of alternatives and decision-makers. Nor is it realistic to think that the researcher knows the complete form of the function $u(\cdot, \cdot)$ mapping attributes into utility values.

McFadden saw that a viable approach does emerge if the researcher has partial knowledge of the form of $u(\cdot, \cdot)$ and partial knowledge of the cross-sectional distribution of the missing attribute data. In particular, he supposed that $u(\cdot, \cdot)$ has the linear-in-parameters separable form

$$u(x_j, s_t) = v(x_{jo}, s_{to}) \cdot \beta + \epsilon_{tj}. \tag{3}$$

Here $v(\cdot, \cdot)$ is a vector-valued function of the components of x_j and s_t that the researcher can observe, labelled x_{jo} and s_{to}. The parameter vector β is to be estimated using the available choice data. The unobserved variable ϵ_{tj} expresses the contribution of unobserved attributes to utility. The cross-sectional distribution of these unobserved variables is assumed known up to the value of a parameter vector α.

With this setup, McFadden derived choice probabilities conditional on the observed attributes. Suppose that a decision-maker is observed to have attributes s_o and to face a choice set C whose alternatives have observed attributes $X_o \equiv (x_{jo}, j \in C)$. Then the random utility model implies that the conditional choice probability for alternative i is

$$P(i|s_o, X_o) = P[v(x_{io}, s_o) \cdot \beta + \epsilon_i \geqslant v(x_{jo}, s_o) \cdot \beta + \epsilon_j, \quad \text{all } j \in C|s_o, X_o, \beta, \alpha]. \tag{4}$$

Subject to identification and regularity conditions, these conditional choice probabilities provide the basis for estimation of the parameters β and α using data on the observed choices and attributes of a random sample of decision-makers. McFadden proposed maximum likelihood estimation of the parameters and studied its statistical properties.

McFadden's development of conditional choice probabilities borrows from

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the long econometric tradition of using linear models to express outcomes of interest. Here it is utility that follows the linear model. The crucial distinction between the present linear model and conventional ones is that the present dependent variable (utility) is unobservable. All that is observable is the identity of the alternative that maximizes utility. Hence equation (4), expressing the conditional choice probability in terms of the latent utilities, becomes the focal point of empirical analysis.

There is some resemblance between McFadden's derivation of conditional choice probabilities from the RUM and Tobin's (1958) derivation of mixed conditional densities and probabilities from censored outcome data, later formalized in Amemiya (1973). The two derivations are sometimes linked under the general rubric of "limited-dependent and qualitative variable" problems; see e.g. Maddala (1983).

(iv) Practicality through the Extreme-value Distribution

At this point McFadden had completed the conceptual development of modern discrete choice analysis. The remaining question was practicality, a matter whose interpretation must depend on the computational power of the day.

McFadden was concerned that maximum likelihood estimation of the parameters (β, α) requires numerous calculations of conditional choice probabilities and iterative search for the parameter value maximizing the likelihood function. He judged that, with the computational technology available at the time of his work, discrete choice analysis would be practical only if the conditional choice probabilities have a quite simple form. With this in mind, he pragmatically searched for a convenient assumption to impose on the cross-sectional distribution of the unobserved attribute variables $(\epsilon_j, j \in C)$.

He found what he was looking for in earlier work of Marschak (1960) and Holman and Marley, cited in Luce and Suppes (1965). These authors had found that if the ϵ variables are assumed to be independently distributed, each according to the Type I extreme-value distribution, then the conditional choice probabilities have the *multinomial logit* form

$$P(i|s_o, X_o) = \exp[v(x_{io}, s_o) \cdot \beta] \Big/ \sum_{j \in C} \exp[v(x_{jo}, s_o) \cdot \beta]. \tag{5}$$

The distribution of $(\epsilon_j, j \in C)$ having been fully specified, there are no α parameters to be estimated here, only the β parameters.

The binary logit model had been prominent in biostatistics since its introduction by Berkson (1944). In the 1960s, versions of the multinomial logit form had been independently investigated by a number of researchers

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in different disciplines, including Theil in econometrics; see Theil (1971). Nevertheless, McFadden's derivation of the model as a description of population behavior consistent with economic theory was entirely new. Moreover, it yielded a more general version of the multinomial logit model than had appeared earlier. For these reasons, McFadden did not use the name multinomial logit to describe his innovation. Instead he called it the *conditional logit model.*

An Extraordinary Synthesis

McFadden [III, 1974] begins with the statement "A fundamental concern of economics is understanding human choice behavior" (p. 105). He describes his objective as follows: "This paper outlines a general procedure for formulating econometric models of population choice behavior from distributions of individual decision rules" (p. 106). At the end of his concluding section describing application of the conditional logit model, he writes "this statistical procedure provides an analog for qualitative dependent variables of the conventional linear statistical model" (p. 134).

Perhaps the most striking characteristic of the article is its smooth progression from a conceptual contribution of great generality to a practical contribution of immediate usefulness. The article is an extraordinary synthesis of basic economic theory, development of a general econometric approach, and embodiment of that approach in a particular form as practical as the conventional linear statistical model.

III. Subsequent Development of Discrete Choice Analysis

"Conditional Logit Analysis of Qualitative Choice Behavior" [III, 1974] was immediately recognized as a paradigmatic breakthrough. This article and McFadden's contemporaneous empirical analyses, [III, 1975a and III, 1976a] and Domencich and McFadden [1975], fundamentally changed the way econometricians and empirical researchers thought about the econometric analysis of individual behavior. In short order, a group of econometricians and empirical researchers were working to generalize the econometric theory and to explore a range of applications. Discrete choice analysis rapidly developed into one of the foundations of modern econometrics.

The present discussion focuses on McFadden's major methodological contributions and briefly describes some related developments, but does not cover empirical studies applying discrete choice analysis. The empirical literature is too vast and diverse for a concise summary to be possible. Only two years after the publication of McFadden [III, 1974], he was already able to cite completed applications to the study of travel demand, migration, the

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demand for consumer durables, the analysis of college-going behavior, occupational choice, and housing location; see McFadden [III, 1976b, Section IV]. Today, a full-length survey would be needed to summarize the applications in any one of these areas. Major bodies of applications have also developed in marketing, telecommunications, industrial organization, and other domains.

Specification and Estimation of Random Utility Models

Throughout the more than quarter century of research since the publication of McFadden [III, 1974], the econometric analysis of discrete choice has retained intact the three basic aspects of McFadden's original work—the random utility model framework, the attribute characterization of alternatives and decision-makers, and the resulting conditional choice probabilities. The particular distributional assumptions imposed in the conditional logit model, however, have been subjected to close scrutiny and many alternative specifications have been studied. This research has several strands.

Flexible Parametric Distributional Assumptions. Although McFadden saw the conditional logit model as a practical tool for empirical analysis, he was very aware of the restrictiveness of this tight functional form as a description of population choice behavior. Early criticisms of the psychological version of the multinomial logit model were divorced from its RUM interpretation and instead emphasized its *independence of irrelevant alternatives* property; see Debreu (1960) and Tversky (1972). Viewing the matter from the perspective of McFadden's RUM derivation, however, it became clear that the crux of the problem was the assumption that the unobserved attribute variables $(\epsilon_j, j \in C)$ are independent and identically distributed.

In the 1970s, a major concern of the econometricians developing discrete choice analysis was to find alternative parametric distributional assumptions that yield computationally tractable conditional choice probabilities and at the same time are sufficiently flexible in their behavioral implications. Some researchers focused attention on the *multinomial probit* specification, wherein $(\epsilon_j, j \in C)$ are assumed to be distributed multivariate normal; see e.g. Daganzo, Bouthelier, and Sheffi (1977) and Hausman and Wise (1978). At the same time, McFadden worked to generalize the conditional logit model. He retained the assumption that each ϵ has an extreme-value distribution but relaxed the assumption that the variables $(\epsilon_j, j \in C)$ are mutually independent. This yielded the *generalized-extreme-value* model introduced in McFadden [II, 1978a] and studied in more depth in McFadden [III, 1981a]. The latter article is notable more broadly for its wide-ranging survey of RUM specifications.

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Estimation Using Simulation Methods. In the 1980s, the computational problem of calculating conditional choice probabilities eased considerably as a consequence of two developments. One, external to the field of discrete choice analysis, was the enormous technological progress in computing manifested at two extremes by the personal computer and the supercomputer. The other, internal to the field of discrete choice analysis, was the development of Monte Carlo simulation approaches to the calculation of choice probabilities and the estimation of discrete choice models.

Lerman and Manski (1981) introduced the idea of approximating conditional choice probabilities at specified parameter values (β, α) by repeatedly drawing pseudo-realizations of $(\epsilon_j, j \in C)$ and, for each realization, determining the alternative that maximizes utility. They suggested that the resulting simulated choice frequencies be used in maximum likelihood estimation of (β, α).

McFadden [III, 1989a] proposed the *method of simulated moments* as an estimation approach. Whereas maximum likelihood estimation using simulated frequencies requires repeated drawings of pseudo-realizations of $(\epsilon_j, j \in C)$ for each decision-maker in the sample, the method of simulated moments yields consistent estimates with only one drawing per decision-maker. Although the method of simulated moments is a beautifully simple idea, the method poses difficult issues in asymptotic statistical analysis and in parameter search. McFadden [III, 1989a] resolved the basic statistical issues. The parameter search issues have been investigated in more recent work. Hajivassiliou (1993) reviews this literature.

Non-parametric Distributional Assumptions. Concern with the restrictiveness of any parametric distributional assumption on the unobserved variables $(\epsilon_j, j \in C)$ has led to the development of a branch of discrete choice analysis that imposes weaker, non-parametric distributional assumptions. The first work of this type was the introduction of the *maximum score* estimation method in Manski (1975) and was followed by the study of non-parametric maximum likelihood estimation by Cosslett (1983). A coherent literature took shape in the mid-1980s and has continued to develop. Horowitz (1993) reviews this literature.

Discrete-continuous Choice Models

The analysis of discrete choice is a relatively new concern for economists. Much older is the econometric analysis of classical demand that deals with the consumer's or firm's selection of quantities for goods available in continuous amounts. A natural step in the evolution of choice analysis was the development of econometric models that explain both discrete and continuous choices.

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Early proposals of *discrete simultaneous equations systems* by Amemiya (1974), Heckman (1978) and others were not derived from well-posed behavioral assumptions, a fact that manifests itself in the internal inconsistencies that these systems can exhibit; see the critique by Schmidt (1981). McFadden [III, 1979] set out to develop an econometric analysis of discrete–continuous choice from the same random-utility-model foundation that he had previously used to study discrete choice problems. This effort continued in McFadden [III, 1981a] and culminated in Dubin and McFadden [VI, 1984a]. Following the same research philosophy as McFadden [III, 1974], the Dubin–McFadden article nicely integrates a general conceptual contribution with a practical contribution of immediate usefulness. Moreover, the article gives an interesting empirical application to a mundane but real discrete–continuous choice problem: the household's discrete choice among differentiated residential electric appliances with varying energy consumption, and its continuous choice of a quantity of energy to consume.

Stratified Sampling Processes

The early development of discrete choice analysis assumed that the available data were observations of the choices made by and attributes of a random sample of decision-makers. In practice samples are often stratified, with the population divided into strata and observations drawn at random within strata. In *exogenous sampling* the survey designer selects decision-makers with given attributes as strata. In *choice-based sampling* the designer selects persons with given choices as strata. The general term *endogenous sampling* is used to decribe any sampling process, such as a choice-based process, that is not based on an exogenous stratification.

Estimation of discrete choice models from data obtained by exogenous sampling poses no intrinsic problems beyond those arising under random sampling. Estimation from data obtained by endogenous sampling, however, does pose new problems. The estimation of discrete choice models from data obtained by endogenous sampling was first analyzed by Manski and Lerman (1977), who focused on the leading case of choice-based sampling and developed a computationally simple estimator with satisfactory statistical properties. Manski and McFadden [III, 1981b] substantially generalized the analysis to cover a broad class of endogenous sampling processes. This article introduced a number of new estimation methods and also examined the problem of optimal sample design. At the same time, Cosslett (1981) studied efficient maximum likelihood estimation under endogenous sampling.

Choice-based sampling has the same formal structure as the *case-control* sampling approach used routinely in epidemiological studies; see Cornfield (1951). Hsieh, Manski and McFadden [VII, 1985] brought the econometrics

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work to the attention of a broad statistical and biostatistical audience. Since then, biostatisticians have used and extended the econometric contributions.

IV. Econometrics and the Nobel Memorial Prize

The Nobel Memorial Prize in Economics has twice been awarded in recognition of early contributions to econometric theory (the inaugural award to Frisch and Tinbergen and the 1989 award to Haavelmo) and once in recognition of contributions to applied econometrics (the 1980 award to Klein). Another recipient of the Memorial Prize, Koopmans, made central contributions to econometrics but was cited for other research. Until this year, award of the prize had yet to recognize any of the major developments of modern econometrics.

It is most gratifying that the Prize Committee has now chosen to honor Daniel McFadden for his research on discrete choice analysis. The econometric analysis of discrete choice has fundamentally changed the way empirical economists study individual behavior. Daniel McFadden made the breakthrough by developing a general procedure for formulating econometric models of population choice behavior from distributions of individual decision rules. He has subsequently fostered the development of discrete choice analysis through his own major contributions and through his stimulation of the work of others. McFadden's research combines conceptual contributions of great generality with practical contributions of immediate usefulness. These surely are the characteristics of a recipient of the Nobel Memorial Prize in Economics.

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